

# A Simple High-Performance Broadband NMR Transmit-Receive System

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A broadband transmit-receive system with excellent isolation, operating in the frequency range 15 to 100 MHz, is described. In transmitting mode the transmitter-probe attenuation for the strong RF pulse does not exceed 1.2 dB, while the transmitter-receiver isolation is better than 42 dB. In receiving mode the transmitter-receiver isolation for the weak nuclear signal is better than 40 dB and the probe-receiver attenuation is equal to or less than 2.6 dB. The system is simple and relatively cheap.

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A well-designed transmit-receive system should fulfill several conditions. In the transmitting mode, the strong RF signal from the transmitter must be delivered to the probe with low attenuation and simultaneously the transmitter-receiver isolation should be as high as possible to avoid damage to the receiver and to keep the gain of the receiver constant and independent of the length of the RF pulse. In the receiving mode, the weak NMR signal is transmitted from the probe to the receiver and therefore the probe-receiver attenuation should be as low as possible. During this mode, the transmitter-receiver isolation should be very high in order to filter noise and low-level RF leakage from the transmitter which could reach the receiver input and distort the NMR signal.

In a previous paper [1] we described a selective transmit-receive system which fulfills all the above-mentioned conditions. The system described in this paper also meets these requirements, but operates over a wide frequency range.

## GENERAL DESCRIPTION OF THE SYSTEM

A diagram of the transmit-receive system is shown in Fig. 1. The ENI power amplifier used in our system is connected to the probe through the RF high-voltage pass (HVP). In transmitting mode, the HVP conducts the strong RF signal with small attenuation (1.2 dB or less). At the same time the receiver is protected against the strong RF signal by means of the protector, connected to the probe and the receiver. The protector has a high input impedance for the strong RF signal, attenuating it more than 42 dB. This means that the strong signal goes only from port 2 of the HVP to port 3 of

the probe. The attenuation of the strong signal by the protector is shown in Fig. 2 as a function of frequency. In the range of frequencies of interest to us, 15 to 100 MHz, the attenuation of the strong signal is larger than 42 dB, implying that, during the transmitting mode, excellent isolation between the transmitter and the receiver is achieved. It would be impossible to obtain such effective isolation with a single  $\lambda/4$  cable terminated by crossed diodes. For example, a  $\lambda/4$  cable for 16.5 MHz terminated by four pairs of 1N4148 crossed diodes gives an attenuation of only 26 dB of the strong signal.

In receiving mode, the HVP strongly attenuates noise and possible leakage from the ENI power amplifier, while at the same time the weak nuclear signal from the probe passes through the protector with negligible attenuation. The attenuation of the weak signal from the HVP is shown in Fig. 3 as a function of frequency. The attenuations at 15, 60, and 100 MHz are 46.5, 44.4, and 40.1 dB, respectively. The frequency dependence of the attenuation of the weak signal by the protector is shown in Fig. 4. The attenuations at 15, 60, and 100 MHz are 1.16, 1.7, and 2.6 dB, respectively.

It should be mentioned that the input impedance for a strong signal at port 4 of the protector (Fig. 1) is high but limited. When port 4 is disconnected from point 0, the signal level at port 3 of the probe is increased slightly, implying that the protector introduces a loss in power delivered to the probe. The loss of power delivered to the probe due to the limited input impedance of the protector for a strong RF pulse is presented in Fig. 5 as a function of frequency. At 15 MHz this value is 1.5 dB; however, in the frequency range 20 to 90 MHz it is less than 1 dB, while at 100 MHz it is 2.5 dB. Below 15 MHz and above 120 MHz the loss of power increases drastically. For example, at 10 and 130 MHz the power loss is 8 dB. This characteristic indicates the useful bandwidth of the protector.

## CIRCUIT DIAGRAMS

*RF high-voltage pass.* The circuit diagram of the HVP is shown in Fig. 6. The high-pass filter  $L_1C_1L_2$  suppresses the spurious low-frequency signal from the ENI transmitter [1].

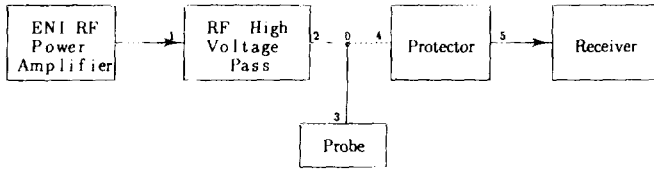


FIG. 1. Diagram of the transmit-receive system.

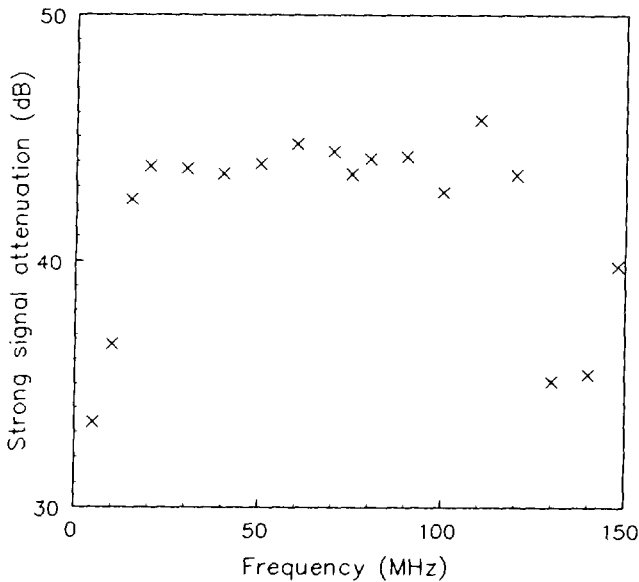


FIG. 2. Attenuation of the strong signal by the protector as a function of frequency.

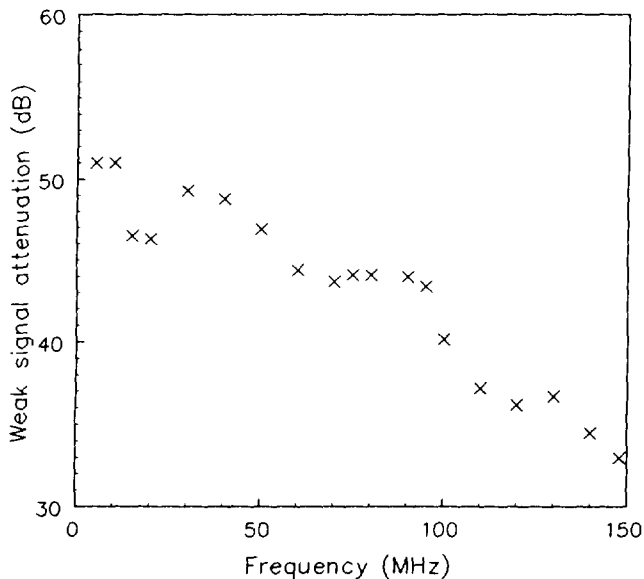


FIG. 3. Attenuation of the weak signal by the RF high-voltage pass as a function of frequency.

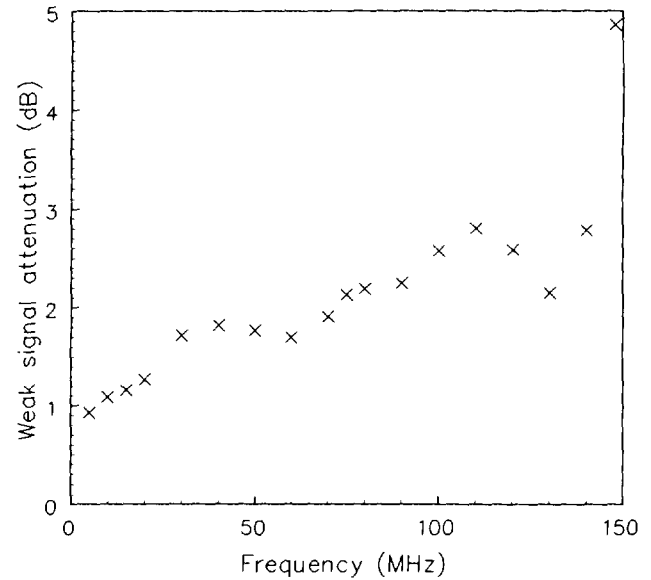


FIG. 4. Attenuation of the weak signal by the protector as a function of frequency.

Transformers  $T_1$  and  $T_2$  operate as broadband  $180^\circ$  phase shifters and therefore it is possible to eliminate leakage and noise passing through the diodes [2]. Carefully adjusting capacitors  $C_3$  and  $C_5$  at high frequency yields maximum attenuation of the weak signal at all frequencies. Each transformer is connected to diodes through small-capacity capacitors  $C_2$ ,  $C_3$ ,  $C_4$ , and  $C_5$  and therefore the loss of power due to the transformers is not observed in the transmitting mode. Each of these transformers has 13 turns bifilar on a ferrite ring. Capacitors  $C_2 = C_4 = 10$  pF and adjustable capacitors  $C_3$  and  $C_5$  have maximum capacity of 10 pF. The

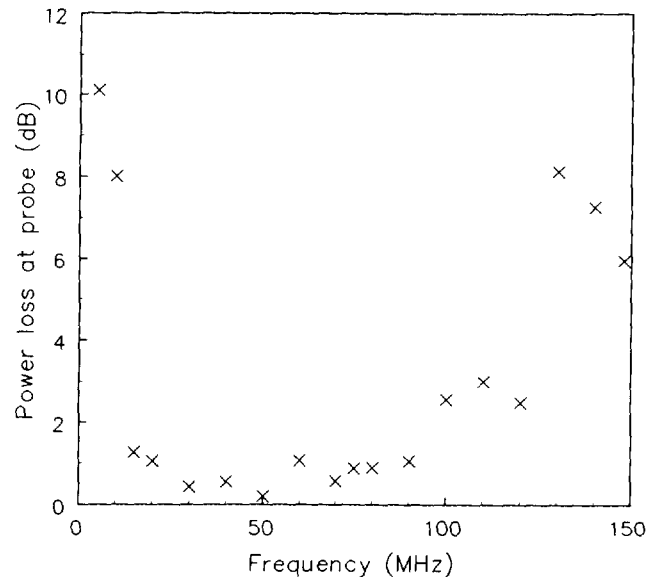


FIG. 5. Power loss at the probe as a function of frequency.

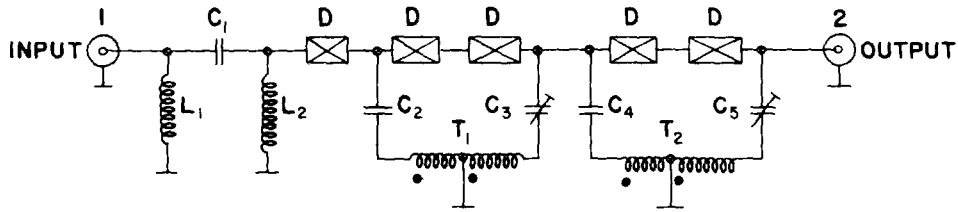


FIG. 6. Circuit diagram of the RF high-voltage pass.

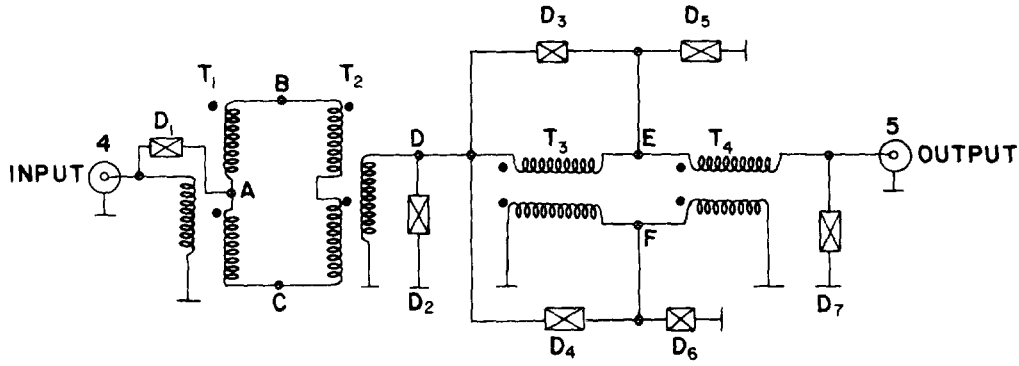


FIG. 7. Circuit diagram of the protector.

high-pass filter was designed for a minimum frequency of 5 MHz ( $L_1 = L_2 = 1.6 \mu\text{H}$ ,  $C_1 = 637 \text{ pF}$ ). Each diode  $D$  consists of four pairs of crossed diodes 1N4148.

**Protector.** The circuit diagram for the protector is shown in Fig. 7. It consists of four broadband transformers and suitably connected crossed diodes. The protector behaves like a  $\lambda/4$  cable terminated by crossed diodes, but it has broadband features. It has a high input impedance for strong signals and a low ( $50 \Omega$ ) input impedance for weak signals.

Let us consider the functioning of the protector. In the receiving mode, none of the diodes conducts and the transformers operate conventionally. A weak signal introduced to port 4 has opposite phases at points B and C and also at points E and F and reaches port 5 with negligible loss in power.

In the transmitting mode for the strong signal, all the diodes in the protector conduct. If a strong signal is applied to port 4, diodes  $D_1$  conduct and the RF is supplied to transformer  $T_1$  mainly at point A. Therefore the RF signal has the same phase at B and C, point D is grounded through diodes  $D_2$ , and transformer  $T_2$  does not operate due to the high input impedance. Port 4 also has a high input impedance since practically no power is transmitted by the protector. Usually the level of signal at D is of the order of 10 Vpp. This signal drops along  $D_3$ – $D_5$  and  $D_4$ – $D_6$ , supplying transformers  $T_3$  and  $T_4$  at E and F with RF signals which are in phase. Therefore transformer  $T_4$  practically does not operate. The result is that the observed signal at port 5 is at a level of less than 2 Vpp.

The protector was tested for power from the ENI amplifier in the range 100 to 200 W. The typical value of the signal level at port 4 was larger than 200 Vpp. Some values of the signal levels at the protector and the attenuations for three frequencies are listed in Table 1. In this table  $U_4$  is the input signal level at port 4,  $U_5$  is the output signal level at port 5, and  $U_D$  is the signal level at point D.

A comparison of the attenuations at D and at port 5 shows that the attenuation of the first pair of transformers is more efficient than that of the next pair. It would be possible to replace transformers  $T_3$  and  $T_4$  by a system identical to transformers  $T_1$  and  $T_2$  that would lead to an attenuation at port 5 which is larger than that listed in Table 1. However, in this case the attenuation of the weak signal in the receiving mode would also be larger. It seems that the system shown in Fig. 7 offers a good compromise for the attenuation of weak and strong signals.

TABLE 1  
Some Signal Levels in the Protector and Attenuations  
as a Function of Frequency

| Frequency<br>(MHz) | $U_4$<br>(V) | $U_D$<br>(V) | $U_5$<br>(V) | Attenuation<br>at D (dB) | Attenuation<br>at port 5 (dB) |
|--------------------|--------------|--------------|--------------|--------------------------|-------------------------------|
| 15                 | 232          | 12.7         | 1.7          | 25.2                     | 42.7                          |
| 60                 | 281          | 8.9          | 1.6          | 30.0                     | 44.9                          |
| 100                | 202          | 4.6          | 1.5          | 32.9                     | 42.6                          |

Note. The symbols are described in the text. All signal levels are peak-to-peak values.

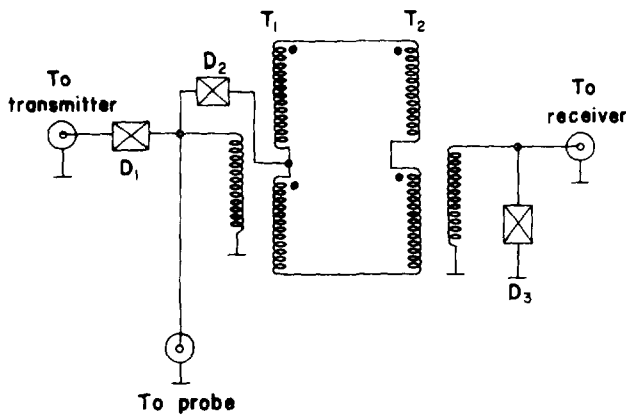


FIG. 8. Broadband duplexer.

Finally we suggest a broadband duplexer, shown in Fig. 8, which could be used with a power amplifier working in class C. The isolation between the transmitter and the receiver in the transmitting mode is not worse than that which could be obtained by a  $\lambda/4$  cable terminated by cross diodes.

*Components of the protector.* The components are as follows:  $D_1$ , two pairs of crossed diodes, 1N4148;  $D_2$ , four pairs of crossed diodes, 1N4148;  $D_3$ ,  $D_4$ ,  $D_5$ , and  $D_6$ , one pair of crossed diodes, 1N4148;  $D_7$ , two pairs of crossed diodes, 1N4148;  $T_1$  and  $T_2$  transformers, trifilar twisted six turns

on ferrite ring;  $T_3$  and  $T_4$  transformers, bifilar twisted six turns on ferrite ring. The protector described functions well within the frequency range 15 to 100 MHz. The upper limit of the range can be increased by decreasing the number of turns on the ferrite rings.

### CONCLUSIONS

The broadband transmit-receive system described in this paper is very simple to construct, has excellent isolation features, and is relatively nonexpensive in comparison with the system reported in the literature [3].

### ACKNOWLEDGMENTS

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